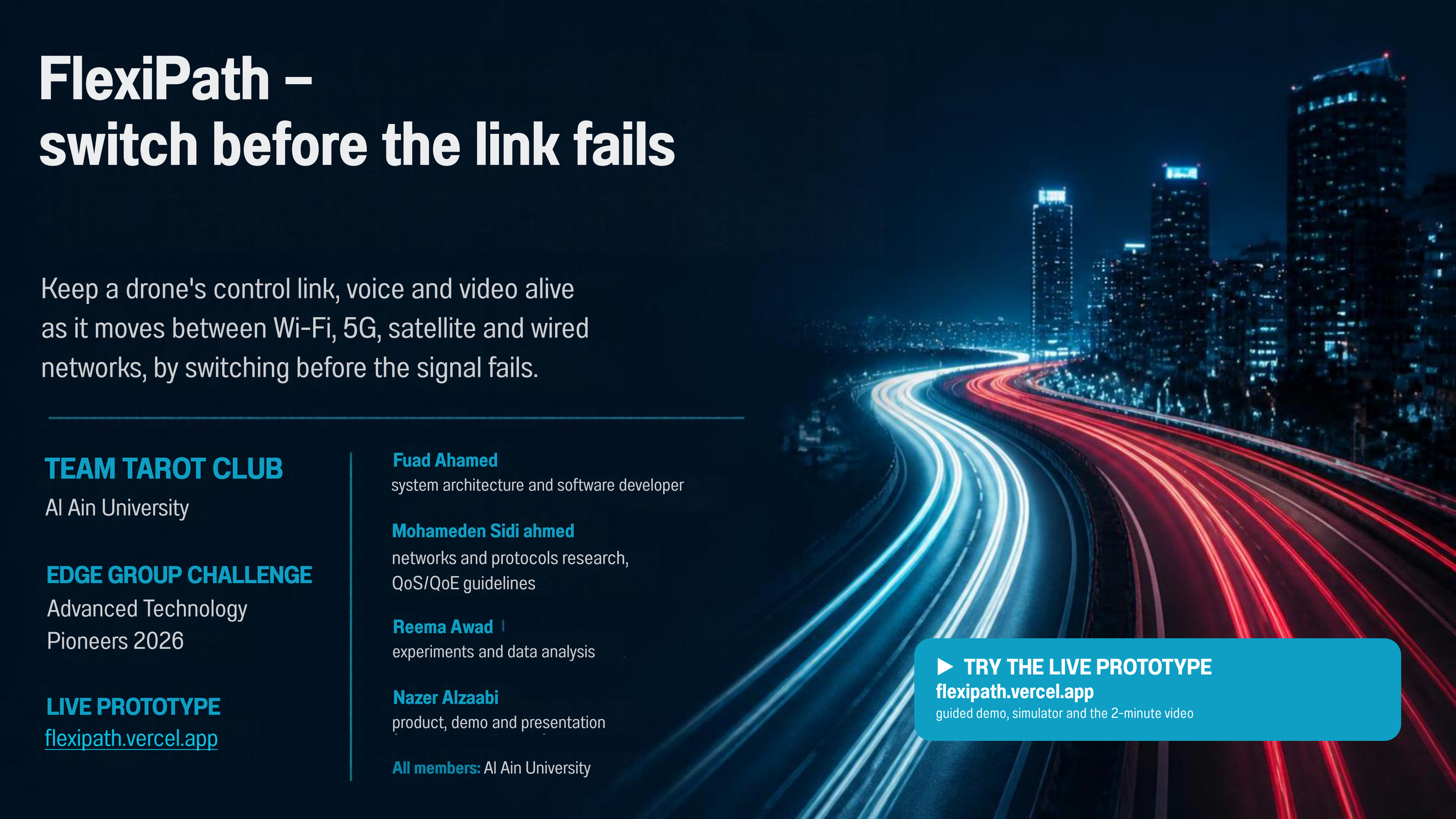
FlexiPath – switch before the link fails

Keep a drone's control link, voice and video alive as it moves between Wi-Fi, 5G, satellite and wired networks, by switching before the signal fails.



TEAM TAROT CLUB
Al Ain University

EDGE GROUP CHALLENGE
Advanced Technology Pioneers 2026

LIVE PROTOTYPE
flexipath.vercel.app

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▶ TRY THE LIVE PROTOTYPE
flexipath.vercel.app
guided demo, simulator and the 2-minute video

Executive summary

FlexiPath keeps critical connections alive without wasting power

PROBLEM

Today's devices drop the old network before they join the next one. A moving user loses the connection, and a drone whose control link (C2) goes silent for more than 300 ms must stop and hover in mid-air (3GPP TS 22.125 rule).

MECHANISM

FlexiPath spots a fading signal seconds early, wakes the next network and tests it, sends only the critical commands on both networks for 0.4 s, then switches with no gap, while a fixed relay keeps every connection open.
Spare radios sleep the rest of the time to save power.

EVIDENCE

160 paired simulated flights (8 scenarios, 5 seeds, 4 speeds) against today's standard stack and the two existing methods: a backup that switches after failure (MPTCP/ATSSS) and always sending twice (PDCP duplication).
FlexiPath kept the control link on 100 % of flights in every scenario; the standard stack on 0 %; the backup method failed in three scenarios; always sending twice uses 2.1–2.6× the power.

DELIVERED

4–7 points more reliability per extra watt for FlexiPath
versus 0.3–0.9 for always sending twice.

A working prototype (interactive 3D digital twin with a guided demo), a deterministic simulation with a headless benchmark, and a documented model with calibrated parameters.
All results are simulated; no live-network measurements are claimed.



The problem and why it matters

Every network change costs today's devices 3.8 s of silence.

Dropped calls. Frozen streams.
300 ms of silence, and a drone hovers.

Emergency response, industrial automation, connected transport and defence can't wait.

Two drones, one route: Wi-Fi → 5G → satellite.



Why drones: 300 ms without a control link forces a hover (3GPP TS 22.125).

1. DOCK (WIRED)
Stable baseline
mission prep

2. WI-FI 6
Local coverage
on the move

3. 5G SA
Wide-area coverage at
speed

4. LEO SATELLITE
Global reach
beyond line of sight

CONTROL LINK (C2)
Command and control

PUSH-TO-TALK VOICE (REAL-TIME)
Clear, low-latency communications

CAMERA UPLOADS (BULK)
High-throughput data transfer

Existing mechanisms we analysed (criterion 01)

REACTIVE MULTIPATH

MPTCP backup subflow (RFC 8684) and 3GPP ATSSS active-standby.

Keeps a backup network on standby; moves only after losses are confirmed and the backup modem wakes up.

0.35 s
gap that breaches the 300 ms C2 budget

1.20 W
low power consumption

ALWAYS DUPLICATE

MPTCP redundant scheduler and PDCP duplication.

Sends every important packet on two networks, all the time.

99–100 %
critical delivery (most reliable)

2.65–3.39 W
high power consumption and 49–100 % duplicate protected bytes

Still fails when both paths die together (a shared radio shadow).

COMMON SHORTCOMINGS

Neither looks ahead.

Ignores the fading signal.

Ignores the known route.

Neither tells a command apart from a video frame.



WHAT WE ADAPT

Keep the **anchored session** (from multipath).
Keep the **duplication** (from PDCP) but confine it to a **400 ms** burst around a predicted failure.
Add **prediction**, **route knowledge** and **traffic classes**.

References: RFC 8684, Multipath TCP v1 (IETF, 2020) · 3GPP TS 23.501 §5.32, ATSSS · 3GPP TS 38.323, PDCP duplication · 3GPP TS 22.125 §7.2, UAV C2 KPIs · full analysis: docs/existing-mechanisms.md

The FlexiPath mechanism

(criterion 02)



DETECT

Measure the primary link ten times a second.
Other modems sleep in DRX at 1.2 W and report signal once per cycle.

PREDICT

Extrapolate the signal trend to where packet loss reaches 15 %.
Arm the loop under 8 s; trigger under 3 s.
Coverage map warns 8 s ahead of a known shadow.

PREPARE

Wake the best spare modem (1.8 W) and test it with tiny 64-byte heartbeats.
Check its delay and loss before it carries anything.

REPLICATE

For a 400 ms burst (3.4 W) send critical traffic, and real-time traffic if its loss leaves budget, on both paths.
The receiver keeps the first copy of each sequence number and discards the late duplicate, so head-of-line delay is 0 ms.
Bulk traffic is never copied, only paced.

HONEST BEHAVIOUR

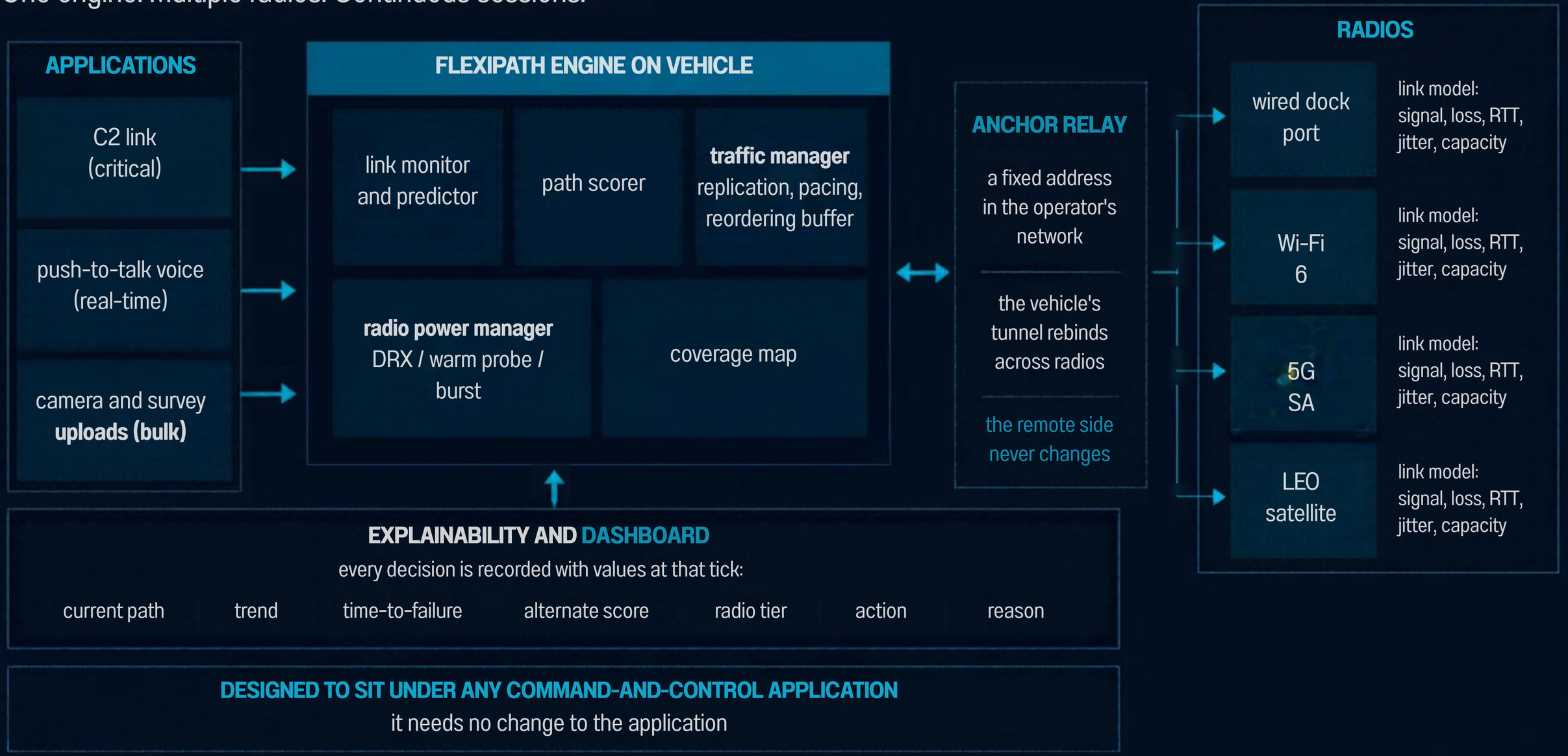
A trend that reverses releases the prediction at no cost.
An unpredictable outage falls back to a fast failover (160 ms in our model).

SWITCH and CONTINUE

Make the tested network the main one with a 0 ms gap and put the old radio back to sleep.
Connections stay open through a tunnel to a fixed relay.

System design

One engine. Multiple radios. Continuous sessions.



Prototype, simulation and model (criterion 03)

flexipath.vercel.app

interactive 3D digital twin with mission and advanced views

guided demo

pauses at each real decision, explains what happened and which panel shows it

deterministic 10 Hz engine

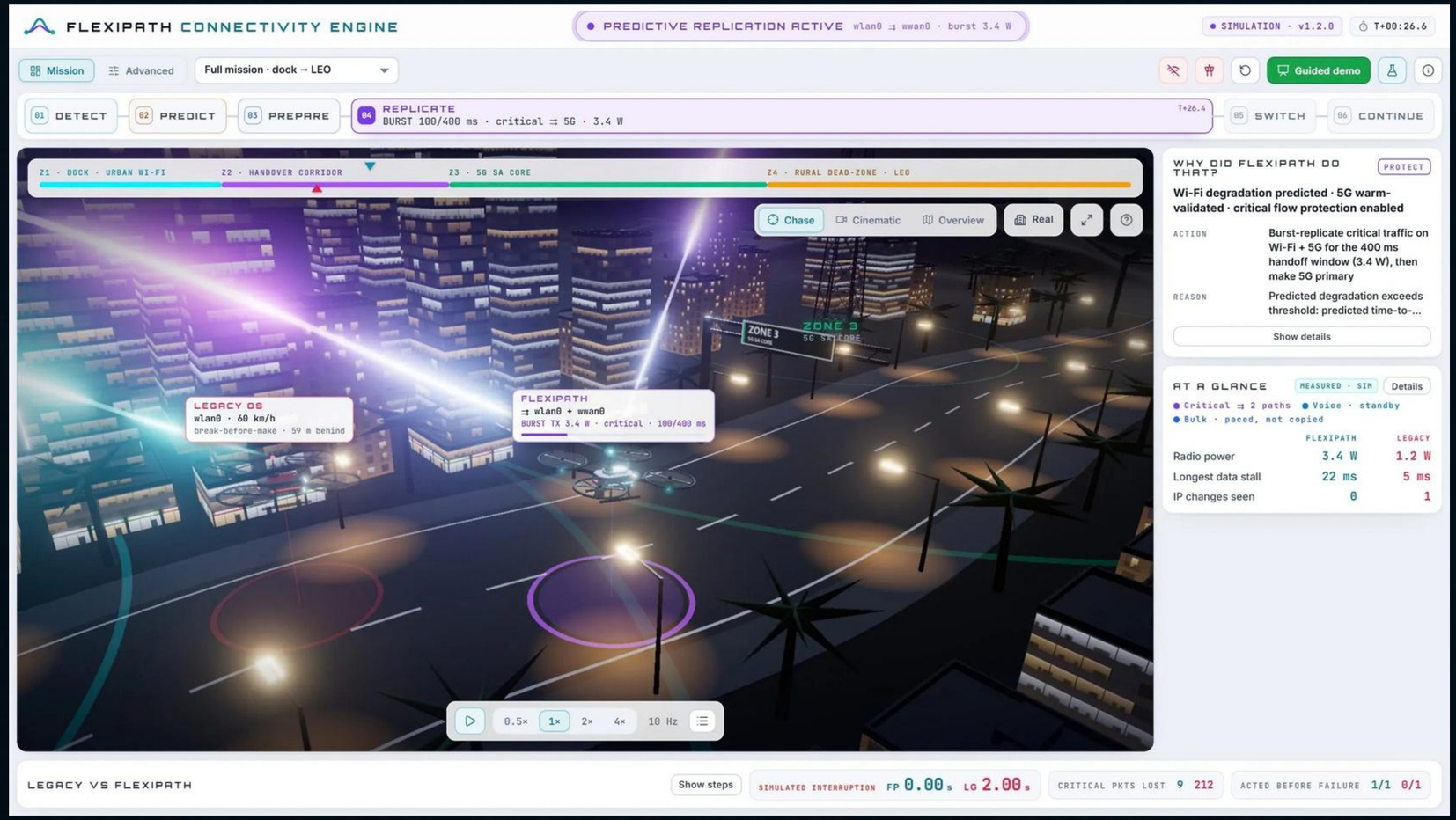
seeded random streams with fading, capacity, loss, RTT, jitter, TCP and radio models

headless benchmark

scenarios A–H × 5 seeds × 4 speeds = 160 paired runs per stack, plus baselines and ablations; exported as CSV with 95 % CIs

published model

parameters checked against published measurements; the 100 ms modem wake tested at 260 ms as a sensitivity run



Mission view of the live prototype: the FlexiPath drone (cyan) and the legacy drone (crimson) fly the same seed and radio conditions, with the decision loop, the decision card and the scoreboard around the twin.

▶ Try it yourself: **flexipath.vercel.app** · the guided demo, the simulator and the 2-minute video

Results against the existing mechanisms (criterion 04)

Paired means over 20 runs per scenario, same seeds for every stack.

100%

FlexiPath kept the drone's control link in every scenario under the piloted 300 ms rule

1.3–4.6 s

FlexiPath acts before the path fails, versus 0.3–0.5 s after for reactive multipath

25–40%

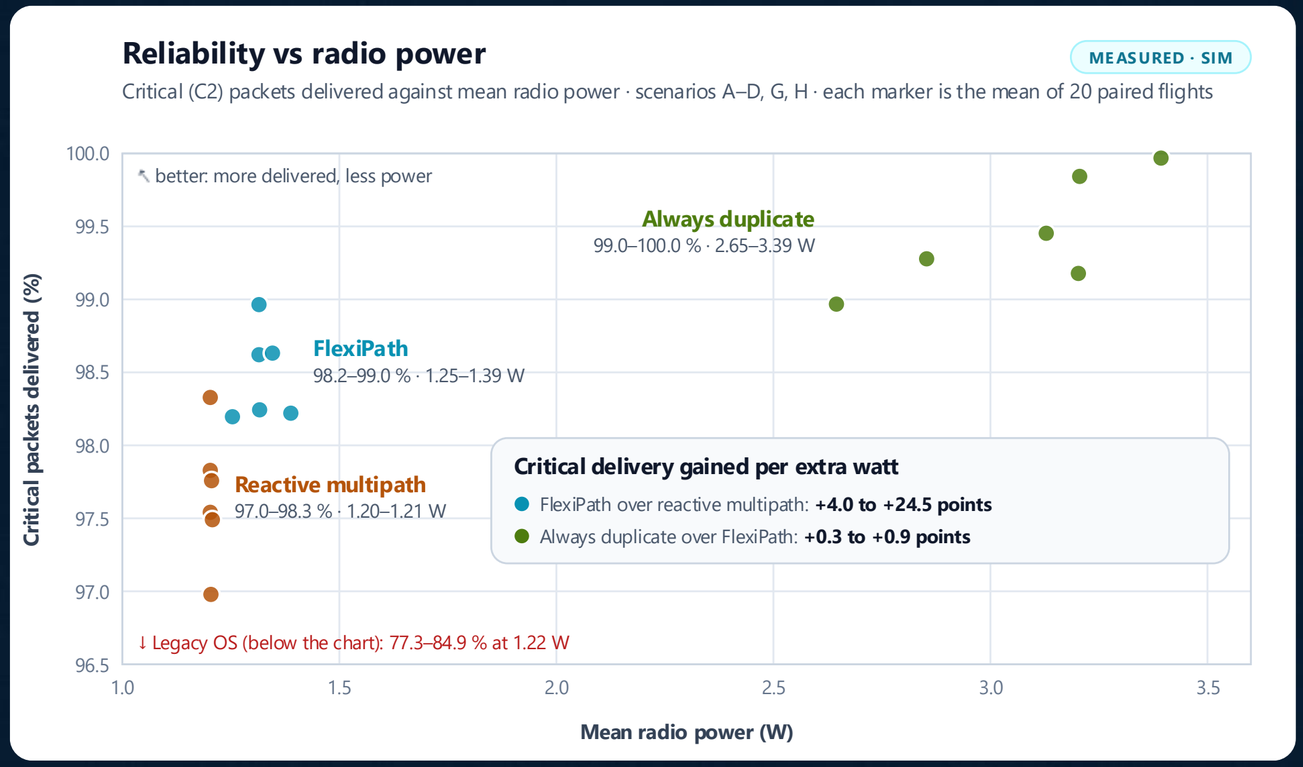
Fewer critical packets lost versus reactive multipath for about 0.1 W more

4.0–7.1 pp/W

Reliability per extra watt versus reactive multipath **(24.5 in D)**

4.5–21×

Less productive: always duplicate over FlexiPath delivers only **0.3–0.9 pp/W**



Reliability against radio power, scenarios A–D, G and H (paired means). FlexiPath sits between the two existing designs: 0.6–1.2 points more critical delivery than reactive multipath for 0.05–0.18 W more, and 2.1–2.6× less power than always duplicate.

Flights that kept C2 under the piloted 300 ms rule (Legacy · Reactive multipath · Always duplicate · FlexiPath): A, B, C 0 · 100 · 100 · 100 — D 0 · 0 · 100 · 100 — G, shared radio shadow 0 · 0 · 40 · 100 — H, map out of date 0 · 0 · 100 · 100.

Robustness: the ablations show trend prediction carries the gain (the burst adds no measurable reliability in-model); the results hold on held-out seeds; under the 1 s waypoint rule every multipath stack keeps C2, so the piloted-rule advantage needs a fast-resume spare modem.

Guidelines for QoS and QoE across heterogeneous networks (criterion 05)

01	Anchor the session, not the interface – a **fixed relay address** removes every reset.
02	Act on the trend, not on the failure – a **3 s signal trend** gives seconds of warning.
03	Prepare the next path just in time, and only then – warm-probe at **1.8 W**, do not keep radios on.
04	Duplicate only what needs it, only around the handover – a **400 ms burst** costs ≈ 1 % of protected bytes.
05	Deliver the **first copy**; never make real-time traffic wait in order.
06	Detect a dead path faster than the application can tolerate – within the **300 ms C2** budget.
07	**Hold still** unless moving clearly pays – a reversed trend releases the prediction at no cost.
08	Measure what **users feel**, on matched pairs – interruption, delivery, MOS, power, on identical seeds.
09	Use what you know about the route – a **coverage map** beats any reaction when two radios die together.



Plan for real-life use, limits, and declaration

Deployment path, proven limits, and our AI-use declaration

▶ TRY THE LIVE PROTOTYPE
flexipath.vercel.app
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DEPLOYMENT

A userspace agent on the vehicle's companion computer (Linux) controls modems via standard interfaces.
The anchor relay sits at the operator's ground station or cloud edge.
It runs under existing C2 and video applications; nothing changes.

LIMITS

All results are simulated.
The radio model is evaluated at ground level.
Piloted rule needs a spare modem that resumes in ~**100** ms (5G RRC_INACTIVE or connected-mode DRX).
At **260 ms**, an unexpected outage still forces a hover.

REAL-LIFE USE

Phase 1 (bench): laptop with Wi-Fi and a 5G dongle runs the included handover probe to measure one real network change. **Phase 2** (field): test drone with Wi-Fi, 5G and a LEO terminal on a surveyed corridor; first survey flights build the coverage map. **Phase 3** (pilot): one operator, one corridor, the paired-comparison method of this study applied to real flights.

AI-USE DECLARATION

AI tools (Anthropic Claude) assisted with code, analysis drafting, this deck and the video; its narration is an AI voice (Kokoro).
The design decisions, the experiments and their interpretation are the team's own, and every member can explain their part in their own words.